Abhishek Prajapati

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SUMMARY

Data Scientist with 7+ years of experience in machine learning, causal inference and experimentation. Skilled in building predictive models, A/B and Quasi testing frameworks, and LLM-based applications using Python, AWS, and SQL. Expert in generative AI landscape, specializing in LLM, LangChain and Agentic AI architectures to automate complex workflows. Experience of working with cross-functional teams to develop, implement and deploy data-driven strategies that improve efficiency and reduce costs.

EXPERIENCE

Project Manager (Data Science - Causal Labs)

The Very Group | EXL **November 2022 – Present, Bengaluru**

- Developed an LLM-powered summarization system using Amazon Bedrock with Llama3 to analyze customer reviews and generate concise, insight-driven summaries
- Designed and implemented Quasi-experimentation frameworks (Causal Impact, Difference in Difference, Pre-Post Variance) enabling measurement where A/B testing was not feasible
- Developed an automated A/B testing platform covering 120+ metrics reducing the analysis time by **48x** (4 Hrs to 5 min)
- Built Triple Exponential Smoothing model to forecast the future Search and Browse conversion based on the intervention and applied a ranged solution using Quasi method. Giving a ranged uplift of 3% (Search conversion),6% (Browse conversion) to the business
- Performed Quasi-testing for MSS using the Difference in Difference technique to estimate the incremental demand, the result shows a significant increase in demand by **£2.7M** over the clearance period
- Explored Prophet model, ARIMA, Bayesian structure time series model for predicting baseline performance, Bayesian outperformed, hence used this to calculate the uplift and estimated ATB-CR uplift as 2% with **95%** confidence interval

Data Analyst

Games & Revenue | MPL **July 2022 – November 2022, Bengaluru**

- Automated daily reporting pipelines using SQL and Python on Databricks
- Built dashboards in Tableau to monitor user behaviour, revenue metrics and matchmaking efficiency
- Improved decision-making speed by delivering real-time insights to stakeholders via Slack integrations

Data Scientist

Analytics & Insights | TCS **October 2018 - July 2022, Bengaluru**

- **Amazon:** Worked on analysis of different kind of loss (Middle Mile Theft) that happens in Amazon, used Redshift and Athena for SQL query and S3 for data storage and alert the DCs
- **The Home Depot: Image content optimization:** Used statistical model, t-test and linear optimization technique to identify the optimal number of images, videos, bullet-points required for a product page based on conversion
- **Fresh Thyme: Basket Impact Model:** Developed a statistical model to predict the sales amount and quantity achieving a model score of **93%**. Also modularized the code script for running the process on CLI, used SQL queries and boto3 to get the data in python script
- **Walgreens: Vendor Compensation Model-** Developed and implemented an On Time In Full and Quantum of Delay model using XGBoost regressor and classifier, identifying the defaulters if **OTIF<90%** and give warning to the vendors for their contract. Streamlined the code running process by modularizing the code, reducing the manual effort for running the script
- **Analytics:** Developed a logistic regression model on diabetic data to identify whether a person is diabetic or not with accuracy of **95%**. Also worked on converting R-script into Python Script

TECHNICAL PROJECTS

- Built a multi-agent AI system for automated report generation, where planner, writer and reviewer agents collaborated to produce structured outputs
- Built an interactive PDF-QA web app using Streamlit, LangChain, and Llama3, enabling real-time querying of unstructured documents
- Developed an LLM-powered summarization tool using Groq API, reducing document length by approx. 70% while preserving key information
- Built a Retrieval-Augmented Generation (RAG) based question-answering system using FAISS as the vector store and Hugging Face Transformers embedding to enhance contextual retrieval

EDUCATION

Integrated Master of Science in Applied Mathematics

Indian Institute of Technology Roorkee **July 2013 – June 2018, Haridwar**

SKILLS

Languages & Python Libraries: Python, SQL, R, Pandas, NumPy, Seaborn, Matplotlib

GenAI & LLMs: LLMs, RAG, LangChain, LangGraph, Hugging Face, Prompt Engineering

Machine Learning: Scikit-learn, TensorFlow, Keras, NLP, Statistical Modeling

Data & Cloud: AWS, Databricks, Teradata, Hadoop, Snowflake

Concepts: Experimentation, A/B testing, MLOps, Causal Inference, Feature Engineering, Data Pipelines, Statistical modeling, Hypothesis testing, Predictive modeling, Forecasting, Data visualization